Repeat purchase prediction project checkpoint

Group 1 | Devano Jonathan | 3036029698 | 5 November 2026

Status. This checkpoint reports Group 1's progress on predicting whether an identified customer will make a merchandise purchase in the next 30 days from the previous 90 days of transactions. The repaired notebook now runs from the checked-in data archive, writes a complete evidence set and passes 19 pipeline contracts. My focus is the text-feature track and the interpretation of the customer-grouped uncertainty intervals.

Project progress

Dataset and design. The project uses the UCI Online Retail dataset (ID 352), covering 541,909 invoice lines from 1 December 2010 to 9 December 2011. The v2 pipeline keeps 391,153 merchandise product lines and separates 1,539 postage, carriage, adjustment, fee and voucher lines. Month-end snapshots from March to October use a 90-day feature window and a 30-day target. March to June trains, July selects, and August to October tests chronologically.

Data audit item	Value	Data audit item	Value
Raw invoice lines	541,909	Missing customer IDs	135,080
Exact duplicate rows	5,268	Missing descriptions	1,454
Merchandise product lines	391,153	Merchandise invoices	18,402
Customer-snapshot rows	16,123	Modelled monthly cohorts	8

Data decision. The one-year dataset remains the main source because every feature and target window is fully covered. Online Retail II adds no useful coverage for these dates: its 2010-2011 sheet is the same target-year data plus one postage line, and its two sheets repeat 22,523 rows from 1-9 December 2010. Exact duplicate removal changes 0.27 percent of valid revenue and none of the eight monthly target labels.

Evidence from the current pipeline

Finding. The repaired LSA-TF-IDF and MiniLM comparisons remove the weak raw sparse-clustering baseline. Logistic behaviour improves over RFM by +0.008 PR-AUC with a paired 95 percent interval of +0.003 to +0.013. The boosting gain is +0.005, but its interval includes zero. Neither MiniLM nor LSA-TF-IDF adds a detectable gain over behavioural boosting.

Model or baseline	July PR-AUC	Aug-Oct PR-AUC
Majority baseline	0.317	0.406
Recency rule	0.407	0.418
Logistic RFM	0.598	0.625
Boosting, behaviour	0.608	0.633
Boosting + LSA-TFIDF	0.615	0.634
Boosting + MiniLM	0.612	0.634

Interpretation. Boosting plus LSA-TF-IDF is the July-selected model, with validation PR-AUC 0.615, final pooled PR-AUC 0.634 and lift at 20 percent of 1.77. The final test does not replace July selection. Paired differences place the leading learned models within overlapping intervals, so the report describes them as indistinguishable rather than claiming a winner from a third-decimal difference.

Individual focus and evidence

Devano Jonathan. My track reviews the text representations and error profile. TF-IDF now uses 100 singular-value components before KMeans; its silhouette rises from 0.015 to 0.145 at K=20. MiniLM reaches 0.075 at K=20. A matched K=16 comparison is included. October error analysis shows that false positives (831 rows, median recency 22.46 days) are more active than false negatives (1,254 rows, median recency 36.51 days), so the model cannot separate every active customer.

Concrete output. The repository now contains the executed notebook, EDA tables, product-cluster examples, cluster-quality tables, temporal contract suite, final predictions, grouped PR-AUC intervals and error tables. Human review of the described clusters and the 90-minute effort estimates remains for each member.

Group contributions

Member	Work completed or in progress	Evidence or output	Approx. hours	Next checkpoint task
Stacia Benita	Data audit, product/charge taxonomy, EDA and duplicate sensitivity.	EDA tables and figures, cleaning audit, duplicate check and dataset comparison.	Confirm	Confirm hours and review the data section.
Devano Jonathan	Reviewed the LSA-TFIDF and MiniLM cluster comparison and October errors.	Cluster-quality tables, examples, error table and limitations draft.	Confirm	Confirm hours and accept cluster descriptions.
Surjadjajadi Clara Natalie	Repaired the pipeline, added uncertainty intervals and executed the contract-tested notebook.	Executed notebook, validation files, grouped intervals, temporal tests and figures.	Confirm	Confirm hours and review the final report.

Hours are not recorded in WORK_SPLIT.md. Each member should replace "Confirm" with their own approximate total before submission.

Next steps and blockers

Target date	Owner	Task
6 Nov 2026	All	Complete individual checkpoint evidence: data sensitivity, cluster and error review, and verified pipeline outputs.
7 Nov 2026	Each member	Submit an individual two-page checkpoint PDF with confirmed hours.
9-14 Nov 2026	Clara; Devano; Stacia	Integrate sensitivity and cluster-count results, finalise figures, and maintain the iteration log.
15-20 Nov 2026	All	Merge individual sections into the assessed report with one consistent argument and caveat set.
21-25 Nov 2026	All	Build slides, rehearse the story and run a final numerical consistency check.

Current blockers. The missing-output blocker is resolved. The remaining blockers are member confirmation of approximate hours, human acceptance of the cluster labels and final course deadlines. No cleaning change now requires a downstream rerun.

AI assistance and checks

AI compared the workbooks, repaired the notebook, generated the v2 evidence and drafted this checkpoint. The archive, temporal windows, product taxonomy, cluster outputs and metric intervals are checked by an automated contract suite with 19 passing tests. Member hours, cluster-label acceptance and submission wording still require human review.